The Kiwi 3D Seed Challenge

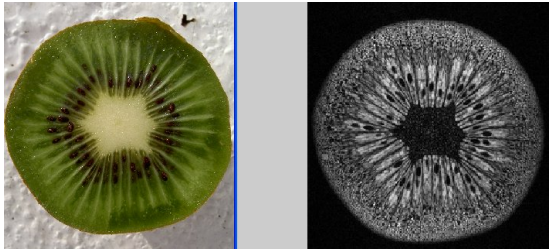
A 6-Week Red Thread in Computational Imaging, Modeling & Semi-Verifiable AI

Arvid Lundervold

Department of Biomedicine & MMIV, University of Bergen
BMED365 — Spring 2027

Lecture 1 Teaser · Monday 4 January 2027

One Intact Kiwifruit, Three Deceptively Simple Questions



Preclinical **7 T Bruker Pharmascan** MRI (UiB, Feb 2005)
$256 \times 256 \times 128$ voxels · $0.234 \times 0.234 \times 0.500$ mm^3

30-Second Classroom Poll

1. Roughly how many seeds (N) are inside this intact kiwifruit? *(200? 600? 1,400? 4,000?)*

2. What fraction of the fruit's 3D volume do the seeds occupy?

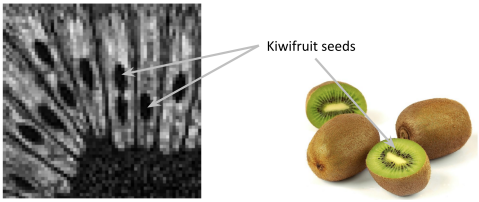
3. The Catch: Nobody ever cut this 2005 fruit open to count the seeds. *How can we verify if an AI or algorithm is right?*

- **Zero medical jargon on Day 1:** Everyone knows what a kiwifruit looks like inside.
- **Real 3D MRI physics:** Anisotropic slices (0.50 mm $>$ 0.234 mm), partial-volume blurring, and closely packed 3D instances.

Why a Kiwifruit in Biomedicine? Quantitative "Lesion Load"

Identical computational task in clinical neuroimaging:

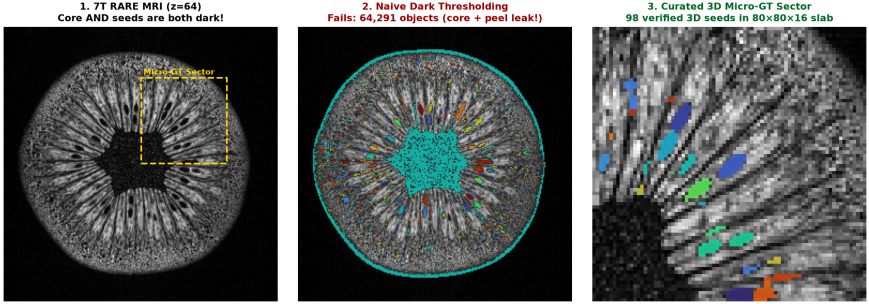
7 T Kiwifruit MRI	Clinical / Biomedical Analogy
Whole fruit ($V_{\text{kiwi}} \approx 86$ cm^3)	Brain parenchyma (skull-stripped MRI)
Dark 3D seeds ($N \approx ?$)	MS plaques, cerebral microbleeds, micrometastases, or 3D cell nuclei
Seed fraction $V_{\text{seeds}}/V_{\text{kiwi}}$	**Quantitative "Lesion Load"**
Touching seeds in a chamber	Confluent vs. discrete focal lesions



MRI Physics Twist: Contrast Inversion!

In an optical photo, the central core (*columella*) is **bright white**. In T_2-weighted RARE MRI (TE = 63.9 ms), dense fibrous core tissue has short T_2 and turns **dark** — mimicking seeds!

Two Day-1 Traps: Naive Thresholding vs. Raw Chatbot Counting



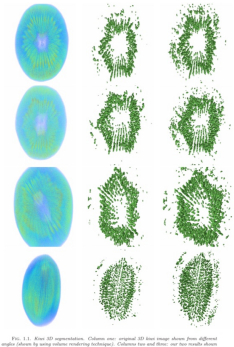
Trap 1: "Just Threshold Dark Voxels"

Darkness $\neq$ seed! Thresholding leaks into the dark central core and outer peel, producing **64,291 fragments** + a giant 11,300 mm^3 megablob.

Trap 2: "Ask a Vision LLM to Count"

Multimodal chatbots output a fluent, confident integer on `kiwi_064.png` — yet hallucinate counts and ignore 128-slice 3D geometry. **Use AI to write & verify 3D code!**

Semi-Verifiability: Auditing Algorithms Without Complete Ground Truth



Dagstuhl / CUHK Milestone (2012): 3D convex Mumford–Shah surfaces by X. Cai & R. H. Chan.

"Could you give us comments as to the quality of the segmentation?"

How our Lab 4 Verification Harness checks any method:

1. **Multi-Planar 3D Overlay:** Axial ($z = 64$) + coronal ($y = 128$) alignment across anisotropic slices.
2. **Botanical Volume Prior:** Real *Actinidia* seeds span 0.35–3.5 mm^3 ($\sim$ 13–127 voxels).
3. **Anatomical Zone Check:** Seeds sit in bright radial locules ($0.25 \leq r/R \leq 0.65$); $\sim 0\%$ in core or peel.
4. **Synthetic 3D Injection:** Implant $K = 25$ known synthetic seeds and verify sensitivity ($\pm 10\%$).
5. **Curated 3D Micro-GT Sector ($80 \times 80 \times 16$):** 98 verified 3D seeds give objective **Precision, Recall, F1** & bias-calibrated whole-fruit count ($\hat{N}_{\text{cal}} \approx 1{,}440$, lesion load $\approx 1.7\%$).

Your 6-Week Red Thread: Same 3D MRI, Evolving AI Paradigms

Course Module	Paradigm on the Kiwi MRI	Key Question
Quick Start & Lab 0 (Wk 1–2)	Voxel features, top-hat filters, K-means / AutoML	What separates seeds from core/peel?
Lab 1: Networks (Wk 1)	3D spatial centroid graph in radial chambers	Can anatomy spot outliers & merges?
Lab 2: Deep Learning (Wk 2)	2D/3D U-Net, StarDist, Cellpose, MedSAM / SAM 2	Do foundation models generalize?
Lab 3: GenAI & Agents (Wk 3)	Prompted vs. verifier-guided coding agents	Can an agent optimize our harness?
Lab 4: Comp. Imaging (Wk 6)	**Live Comparative Showdown (Notebooks 01–03)**	Which paradigm wins — and *why*?

Explore Anytime & Optional Block 2 Individual Project

- **Repository & 1-Click Colab Notebooks:** `Lab4-Comp-Imag/Kiwi-challenge` (01, 02, and 03 + curated 3D Micro-GT NIfTI for ITK-SNAP / 3D Slicer).
- **Individual Project Idea (Weeks 4–6):** Benchmark classical, foundation-model (`SAM 2` / `Cellpose`), or agentic 3D segmentation pipelines on this self-contained 7 T dataset!